



VIGNAN INSTITUTE OF TECHNOLOGY AND SCIENCE

Near Ramoji Film City, Deshmukhi Village, Pochampally Mandal, Yadadri Bhuvanagiri Dist.

(Approved by AICTE, New Delhi, Affiliated to JNTUH, Hyderabad)

AN AUTONOMOUS INSTITUTION



Trust - Aware Quantum - Assisted Security System for IoT Network

A MINI PROJECT REPORT

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Assistant Professor

in partial fulfillment for the award of the degree of

BACHELOR OF TECHNOLOGY

in

INFORMATION TECHNOLOGY



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DEPARTMENT OF INFORMATION TECHNOLOGY

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The department aims to produce globally competent graduates to meet the ever changing needs of IT industry through contemporary knowledge and moral values.

MISSION

1. To provide a holistic learning environment to students for professional development and ethical practices.
2. To train students to be industry ready and to have multidisciplinary skills as an individual and as a team.
3. To impart skills beyond university prescribed to transform students into a well-rounded IT professional.
4. To continuously engage in research and projects development, use of emerging technologies to attain self-sustainability.



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DEPARTMENT OF INFORMATION TECHNOLOGY

CERTIFICATE

This is to certify that the thesis work titled — “**Trust-Aware Quantum-Assisted Security System for IoT Network**” submitted by ADDULA SONY (Regd.No. 23891A1202), SIDI KHUSHI (Regd.No.23891A1257), AITHANABOINA POOJITHA (Regd.No. 23891A1204), GANJI SAMPATH KUMAR (Regd.No. 23891A1223) in partial fulfillment of the requirements for the award of the degree of **Bachelor of Technology in Information Technology** to the Vignan Institute of Technology And Science, Deshmukhi is a record of bonafide work carried out by us under my guidance and supervision.

The results embodied in this project report have not been submitted in any university for the award of any degree and the results are achieved satisfactorily.

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Associate Professor

GUIDE

Prof. B.V. Chowdary

Professor & Head of the Department, IT

External Examiner

DECLARATION

We hereby declare that project entitled — Trust-Aware Quantum-Assisted Security System for IoT Network is bonafide work duly completed by us. It does not contain any part of the project or thesis submitted by any other candidate to this or any other institute of the university.

All such materials that have been obtained from other sources have been duly acknowledged.

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ABSTRACT

The rapid growth of Internet of Things (IoT) devices has introduced significant security challenges due to the lack of continuous monitoring and trust evaluation mechanisms. Traditional IoT security systems mainly rely on authentication and encryption, which are not sufficient to detect malicious or compromised devices after they gain access to the network. To address this limitation, this project proposes a Trust-Aware IoT Security System integrated with Edge Computing and Quantum-Assisted Security techniques.

The system focuses on evaluating the behavior of IoT devices in real time by analyzing communication parameters such as packet delivery ratio, transmission frequency, error rate, and response time. A trust score is calculated for each device using a weighted formula, allowing the system to classify devices as trusted or untrusted. The implementation uses an edge server to perform trust evaluation locally, reducing latency and improving response time.

To enhance data security, a quantum-inspired encryption mechanism is incorporated for secure communication between devices and servers. The system is developed using a modular architecture consisting of a simulated IoT device layer, an edge processing layer, and a centralized monitoring server. A web-based dashboard is used to visualize device status and trust scores in real time.

The proposed system improves IoT network security by enabling continuous monitoring, early detection of malicious devices, and efficient data processing. It is scalable, flexible, and suitable for real-world IoT applications where secure and reliable communication is essential.

TABLE OF CONTENTS

ACKNOWLEDGEMENT.....	III
ABSTRACT.....	IV
1. INTRODUCTION.....	1
2. LITERATURE SURVEY.....	2
3. SYSTEM ANALYSIS.....	8
Problem Statement.....	8
Exisiting System.....	9
Disadvantages of Exisiting System.....	9
Proposed System.....	10
Advantages of Proposed System.....	10
Software Requirement Specification.....	11
Hardware Requirements.....	11
Software Requirements.....	11
4. SYSTEM DESIGN.....	12
Introduction.....	12
Data Flow Diagrams.....	12
UML Diagrams.....	15
Class Diagram.....	15
Use Case Diagram.....	18
Sequence Diagram.....	20
Activity Diagram.....	23
System Architecture.....	27
5. SYSTEM IMPLEMENTATION.....	28
Data Acquisition.....	28
Preprocessing.....	29
Decision Making.....	30
Modules.....	31
Monitoring and Alert System.....	33
6. RESULTS & DISCUSSION.....	34

7. SYSTEM TEST.....	37
8. MECHANISMS AND TECHNIQUES.....	40
9. CONCLUSION.....	42
10. FUTURE ENHANCEMENT.....	43
REFERENCES.....	44

LIST OF TABLES AND FIGURES

Fig 4.1: Data Flow Diagram.....	14
Fig 4.2: Class Diagram.....	17
Fig 4.3: Use case Diagram.....	19
Fig 4.4: Sequence Diagram.....	22
Fig 4.5: Activity Diagram.....	26
Fig 4.6: System Architecture.....	27
Fig 6.1: style.css.....	34
Fig 6.2: index.html.....	34
Fig 6.3: app.py.....	34
Fig 6.4: device_simulator.py.....	35
Fig 6.5: server.py.....	35
Fig 6.6: requirements.txt.....	35
Fig 6.7: Terminal 1.....	35
Fig 6.8: Terminal 2.....	36
Fig 6.9: Terminal 3.....	36
Fig 6.10: output.....	36

1. INTRODUCTION

The Internet of Things (IoT) has become an essential part of modern technology, connecting a wide range of devices such as sensors, smart appliances, industrial machines, and embedded systems to the internet. These devices communicate with each other and with centralized servers to perform tasks automatically without human intervention. Applications of IoT can be seen in smart homes, healthcare systems, agriculture, transportation, and industrial automation.

Despite its advantages, IoT introduces significant security challenges. Most IoT devices are designed with limited computational power and memory, which restricts the implementation of complex security mechanisms. As a result, many devices rely on basic authentication and encryption techniques, which are not sufficient to protect against modern cyber threats. Attackers can exploit vulnerabilities in these devices to gain unauthorized access, manipulate data, or disrupt services.

One of the major limitations of traditional IoT security systems is that they focus only on initial authentication. Once a device is authenticated, it is generally allowed to communicate freely within the network. This creates a serious risk because a malicious or compromised device can continue to operate without being detected. Therefore, there is a need for a system that not only authenticates devices but also continuously monitors their behavior.

To address this issue, this project proposes a Trust-Aware Quantum-Assisted Security System for IoT Networks. The main idea behind this system is to evaluate the trustworthiness of each IoT device based on its behavior and communication patterns. Instead of relying only on authentication, the system continuously analyzes device activity using multiple parameters.

The parameters used for trust evaluation include:

- Packet Rate
- Authentication Score
- Anomaly Score
- Communication Behavior

These parameters provide a comprehensive view of how a device behaves within the network. For example, a sudden increase in packet rate or abnormal communication patterns may indicate malicious activity. By analyzing these factors, the system calculates a trust score for each device.

In addition to trust evaluation, the system incorporates a quantum-assisted key generation mechanism. Although full-scale quantum cryptography requires specialized hardware, this project simulates the concept by generating highly secure random keys using cryptographic techniques. These keys are used to secure communication between IoT devices and the edge server.

Additionally, a simulated quantum key generation mechanism is used to enhance communication security between IoT devices and the edge server. Based on the trust score, the system classifies devices into three categories:

- Allow
- Monitor
- Block

This approach helps in identifying suspicious devices and improving overall network security.

2. LITERATURE SURVEY

1. Title: The Internet of Things: A Survey

Authors: Luigi Atzori, Antonio Iera, Giacomo Morabito

Year: 2010

Introduction:

This paper provides a comprehensive overview of IoT architecture, communication models, and challenges. It highlights that IoT devices are highly resource-constrained and often lack strong security mechanisms, making them vulnerable to various cyber attacks.

Methodology:

The authors analyzed different IoT communication technologies, protocols, and system architectures. They also discussed security issues related to device authentication, data transmission, and network management.

Findings:

The study concludes that IoT systems require stronger and more adaptive security solutions beyond traditional authentication and encryption. It emphasizes the need for intelligent mechanisms to monitor device behavior continuously.

Pros:

- Strong foundation for understanding IoT systems
- Identifies key security challenges
- Useful for designing secure IoT architectures

Cons:

- Does not provide implementation-level solutions
- Limited focus on trust-based mechanisms
- Lacks dynamic security models

2. Title: Security and Privacy in Internet of Things: A Survey

Authors: Simone Sicari, Alessandra Rizzardi, Luigi Alfredo Grieco

Year: 2015

Introduction:

This paper focuses on major security and privacy issues in IoT systems. It highlights that existing security solutions mainly rely on authentication and encryption, which are not sufficient for dynamic IoT environments.

Methodology:

The authors reviewed various security mechanisms including access control, data protection, and trust management systems. They analyzed how these mechanisms perform in real-world IoT scenarios.

Findings:

The study emphasizes that **trust management systems** are essential for improving IoT security. Continuous monitoring of device behavior can help detect malicious activities even after authentication.

Pros:

- Detailed explanation of IoT security challenges
- Highlights importance of trust evaluation
- Covers multiple security aspects

Cons:

- Limited practical implementation
- Does not integrate trust with advanced cryptography
- Scalability issues not fully addressed

3. Title: Trust Management in Internet of Things: A Survey

Authors: Zheng Yan, Peng Zhang, Athanasios V. Vasilakos

Year: 2014

Introduction:

Trust management is a key component in ensuring reliable communication in IoT networks. This paper discusses how trust can be used to evaluate device reliability based on behavior and interaction history.

Methodology:

The authors analyzed different trust models that use parameters such as communication patterns, packet transmission rate, and historical interactions. These models were compared based on efficiency and scalability.

Findings:

The study shows that trust-based systems are effective in identifying malicious or unreliable devices. However, most existing models lack real-time decision-making capabilities and efficient integration with security mechanisms.

Pros:

- Provides strong foundation for trust-based systems
- Explains different trust parameters
- Useful for designing trust evaluation models

Cons:

- High computational complexity
- Limited real-time implementation
- Lack of integration with secure communication methods

4. Title: The Security of Practical Quantum Key Distribution

Authors: Valerio Scarani, Hoi-Kwong Lo, Norbert Lütkenhaus

Year: 2009

Introduction:

This paper introduces quantum key distribution (QKD), which provides highly secure communication using principles of quantum mechanics. It explains how quantum keys can prevent interception and unauthorized access.

Methodology:

The authors analyzed different quantum cryptographic techniques and compared them with classical encryption methods. They also discussed implementation challenges and security advantages.

Findings:

Quantum-based key generation offers stronger security than traditional encryption methods. However, practical implementation requires advanced hardware and infrastructure.

Pros:

- Provides very high security
- Resistant to modern cyber attacks
- Detects eavesdropping attempts

Cons:

- Expensive and complex implementation
- Requires specialized hardware
- Not directly applicable to basic IoT systems

5. Title: Edge Computing for the Internet of Things

Authors: Weisong Shi, Jie Cao, Quan Zhang

Year: 2016

Introduction:

Edge computing improves IoT performance by processing data closer to the devices instead of sending everything to the cloud. It also enhances security by enabling faster decision-making.

Methodology:

The authors studied how edge servers can be used for local data processing, anomaly detection, and security enforcement in IoT networks.

Findings:

The study shows that edge computing enables real-time monitoring and reduces response time. It is effective for implementing trust evaluation and security mechanisms.

Pros:

- Low latency processing
- Real-time decision making
- Improves system efficiency

Cons:

- Limited computational resources
- Requires efficient design
- Security depends on edge server reliability

3. SYSTEM ANALYSIS

Problem Statement :

IoT networks consist of a large number of interconnected devices that continuously exchange data. These devices often operate in open and distributed environments, making them vulnerable to various types of cyber attacks. Due to limited security capabilities, IoT devices are easy targets for attackers who aim to gain unauthorized access or disrupt network operations.

Traditional security mechanisms focus mainly on authentication and encryption. While these methods can prevent unauthorized entry to some extent, they do not ensure that a device behaves correctly after authentication. A compromised device can still operate within the network and perform malicious activities without being detected.

Another challenge is the dynamic nature of IoT networks. Devices frequently join and leave the network, and their behavior may change over time. Static security policies are not effective in such environments. Therefore, there is a need for a system that can continuously evaluate device behavior and detect anomalies in real time.

The problem addressed in this project is the development of a system that can:

- ✓ Evaluate the trustworthiness of IoT devices dynamically
- ✓ Detect suspicious or malicious behavior
- ✓ Secure communication between devices and servers
- ✓ Automatically classify devices based on their trust level

Objectives :

The main objectives of the proposed system are:

- ✓ To develop a secure IoT environment that can detect malicious devices dynamically
- ✓ To implement a trust-based mechanism for continuous monitoring of device behavior
- ✓ To enhance communication security using quantum-assisted key generation
- ✓ To reduce response time using edge computing instead of cloud dependency
- ✓ To prevent unauthorized access and data breaches in IoT networks
- ✓ To design a system that is scalable and adaptable to different IoT applications
- ✓ To improve overall network reliability and performance

Existing System :

Existing IoT security systems primarily rely on two mechanisms:

1. **Authentication** – Verifies the identity of a device before allowing access
2. **Encryption** – Protects data during transmission

These methods are widely used because they are simple and efficient. However, they have significant limitations. Authentication ensures that only authorized devices can access the network, but it does not monitor their behavior after access is granted.

For example, if a device is compromised after authentication, it can continue to send malicious data without being detected. Similarly, encryption protects data from being intercepted, but it does not prevent attacks originating from within the network.

Some advanced systems use rule-based monitoring, where predefined rules are used to detect suspicious activities. However, these systems are not flexible and cannot adapt to new or unknown attack patterns.

Disadvantages of Existing System :

The existing system suffers from several limitations:

- **No continuous monitoring:** Devices are not monitored after authentication
- **Vulnerable to insider attacks:** Malicious devices can remain undetected
- **High latency:** Cloud-based processing delays threat detection
- **Limited scalability:** Not suitable for large IoT networks
- **Weak security model:** Relies only on encryption and authentication
- **No trust evaluation:** Cannot differentiate between normal and abnormal device behavior
- **Delayed response:** Slow reaction to security threats
- **Single-layer security:** Lack of multi-level protection

These limitations make existing IoT systems highly vulnerable to modern cyber attacks.

Proposed System :

The proposed system introduces a trust-based approach combined with quantum-assisted security mechanisms to improve IoT network protection.

In this system, each device is evaluated based on multiple behavioral parameters. These parameters are analyzed to calculate a trust score, which represents the reliability of the device.

The system architecture includes the following components:

- IoT Device
- User Interface
- Edge Server
- Trust Calculation Module
- Quantum Key Generation Module
- Decision Module
- Device History Storage

When a device sends data to the server, the system first validates the input parameters. Then, the trust calculation module computes the trust score using a predefined formula. At the same time, the quantum key generation module generates a secure key for communication.

Based on the trust score, the decision module classifies the device into one of three categories: Allow, Monitor, or Block. The result is displayed to the user and stored in the system for future reference.

Advantages of Proposed System :

The proposed system provides several improvements:

- **Continuous monitoring:** Tracks device behavior in real-time
- **Improved security:** Combines trust evaluation with encryption
- **Fast response:** Edge computing reduces delay
- **Detection of malicious devices:** Identifies abnormal behavior quickly
- **Multi-layer protection:** Integrates trust + encryption + edge processing
- **Scalability:** Can handle large IoT networks
- **Efficient resource usage:** Reduces cloud dependency

- **Reliable communication:** Secure data transfer using advanced keys

Software Requirement Specification

Software Requirements :

- Operating System : Windows / Linux
- Programming Language : Python
- Frontend : HTML, CSS, JavaScript
- Backend Framework : Flask
- Database : SQLite / JSON-based storage
- Development Tools : VS Code / PyCharm
- Browser : Chrome / Edge

Hardware Requirements :

- Processor : Intel i3 or above
- RAM : 4 GB
- Storage : 500 GB HDD / SSD
- Internet connectivity

4. SYSTEM DESIGN

Introduction :

System design is an important phase in the development of the proposed Trust-Aware Quantum-Assisted IoT Security System, as it defines how the entire system is organized and how each component interacts to achieve secure and efficient operation. It provides a clear blueprint for implementing the system by describing the structure, data flow, and interaction between modules.

The primary focus of this design is to address the security limitations of traditional IoT systems by incorporating continuous trust evaluation, secure communication mechanisms, and real-time data processing. Unlike conventional systems that rely only on authentication, the proposed design ensures that every device is continuously monitored throughout its operation in the network.

The system follows a modular architecture, where each module is responsible for a specific task. Each module is interconnected in such a way that data flows smoothly from one stage to another, ensuring efficient processing and quick response to any abnormal activity.

Another important aspect of the system design is the use of edge computing. Instead of sending all data to a centralized cloud server, the proposed system processes data at the edge (near the devices). This reduces latency, improves response time, and allows faster detection of malicious behavior.

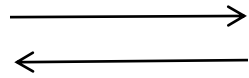
The design also considers scalability, allowing new IoT devices to be added without affecting system performance. Additionally, the modular structure makes it easy to update or modify individual components without redesigning the entire system.

Data Flow Diagram :

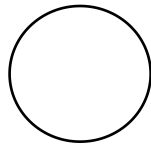
A Data Flow Diagram (DFD) is a graphical tool utilized for describing and analyzing the flow of data within a system, whether manual or automated. It depicts processes, data stores, and data movement throughout the system, illustrating the transformation of input data into output. DFDs serve as the central model from which other system components are developed, providing a logical representation of data movement that is independent of physical system components. They are essential for clearly articulating system requirements.

and serve as the foundation for creating system structure charts during the design phase. The basic notation used in DFDs includes symbols such as processes, data stores, data flows, and external entities, allowing for a visual representation of data movement and system interactions.

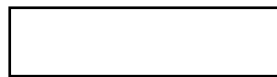
Dataflow: Data move in a specific direction from an origin to a destination.



Process: People, procedures, or devices that use or produce (Transform) Data. The physical component is not identified.



Source: External sources or destination of data, which may be People, programs, organizations or other entities.



Data Store: Here data are stored or referenced by a process in the System.



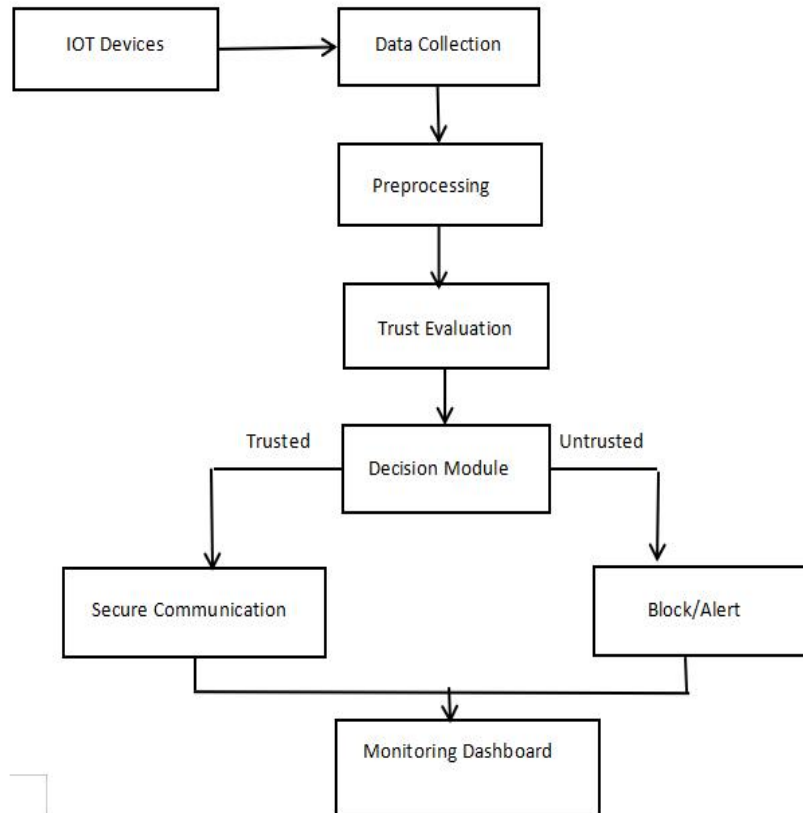


Fig 4.1: Data Flow Diagram of Trust-Aware Quantum-Assisted Security System for IoT Network.

UML Diagrams:

Class Diagram:

Class Diagram gives an overview of a software system by displaying classes, attributes, operations, and their relationships. This Diagram includes the class name, attributes, and operation in separate designated compartments.

Class diagram is a type of static structure diagram that describes the structure of a system by showing the system's classes, their attributes, operations and the relationship among the objects.

Purpose

Shows static structure of classifiers in a system

Helpful for developers and other team members too

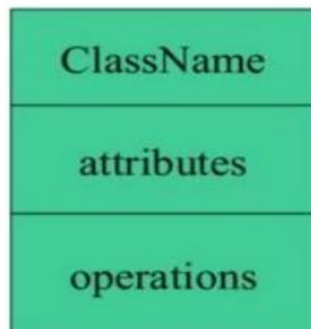
Business analysts can use class diagram to model systems from a business perspective.

Essential elements of A UML class diagram:

Class Name

Attributes

Operations



Class Name:

The name of the class is only needed in the graphical representation of the class. It appears in the topmost compartment. A class is the blueprint of an object which can share the same relationships, attributes, operations, & semantics. The class is rendered as a rectangle, including its name, attributes, and operations in separate compartments.

Following rules must be taken care of while representing a class:

1. A class name should always start with a capital letter.
2. A class name should always be in the center of the first compartment.
3. A class name should always be written in **bold** format.
4. An abstract class name should be written in italics format.

Attributes:

An attribute is named property of a class which describes the object being modeled. In the class diagram, this component is placed just below the name-compartment. A derived attribute is computed from other attributes.

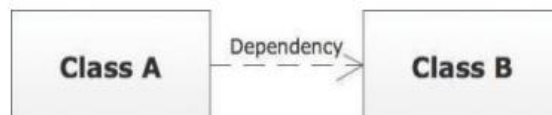
Relationships:

There are mainly three kinds of relationships in UML:

- Dependencies
- Generalizations
- Associations

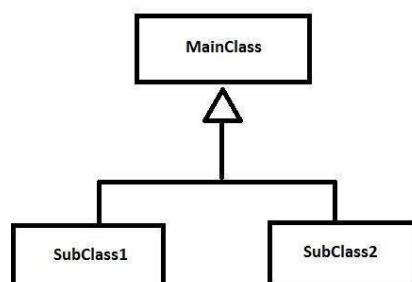
Dependency:

A dependency means the relation between two or more classes in which a change in one may force changes in the other. However, it will always create a weaker relationship. Dependency indicates that one class depends on another.



Generalization:

A generalization helps to connect a subclass to its superclass. A sub-class is inherited from its superclass. Generalization relationship can't be used to model interface implementation. Class diagram allows inheriting from multiple super classes.



Association:

This kind of relationship represents static relationships between classes A and B.

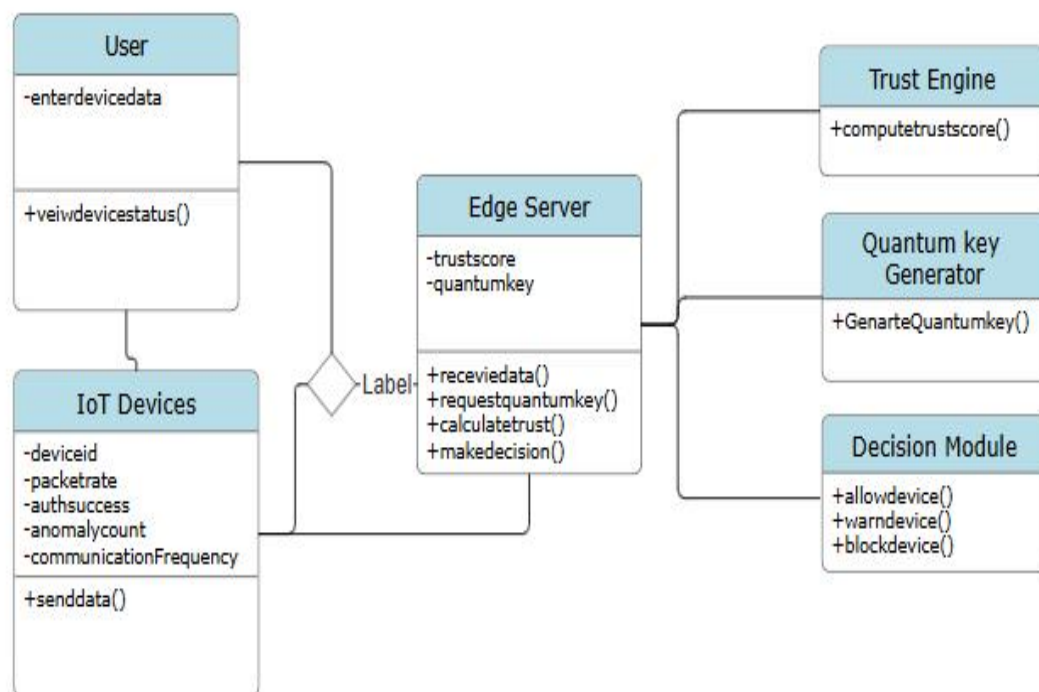
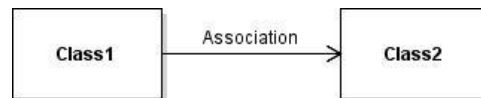


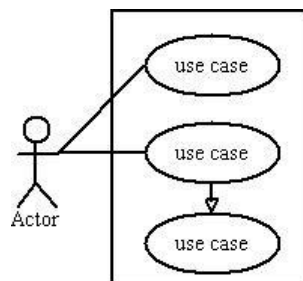
Fig 4.2: Class Diagram of Trust-Aware Quantum-Assisted Security System for IoT Network.

Use Case Diagram:

Use case diagrams model the functionality of a system using actors and use cases. Use cases are services or functions provided by the system to its users. Use case diagram is useful for representing the functional requirements of the system using various notations like system, use case, actors, and relationships.

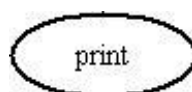
Basic Use Case Diagram Symbols and Notations System:

Draw your system's boundaries using a rectangle that contains use cases. Place actors outside the system's boundaries.



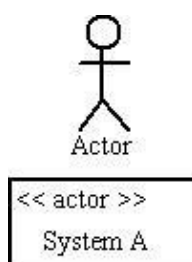
Use Case:

Draw use cases using ovals. Label with ovals with verbs that represent the system's functions.



Actors :

Actors are the users of a system. When one system is the actor of another system, label the actor system with the actor stereotype.



Relationships:

Illustrate relationships between an actor and a use case with a simple line. For relationships among use cases, use arrows labelled either "uses" or "extends." A "uses" relationship indicates that one use case is needed by another in order to perform a task. An "extends" relationship indicates alternative options under a certain use case.

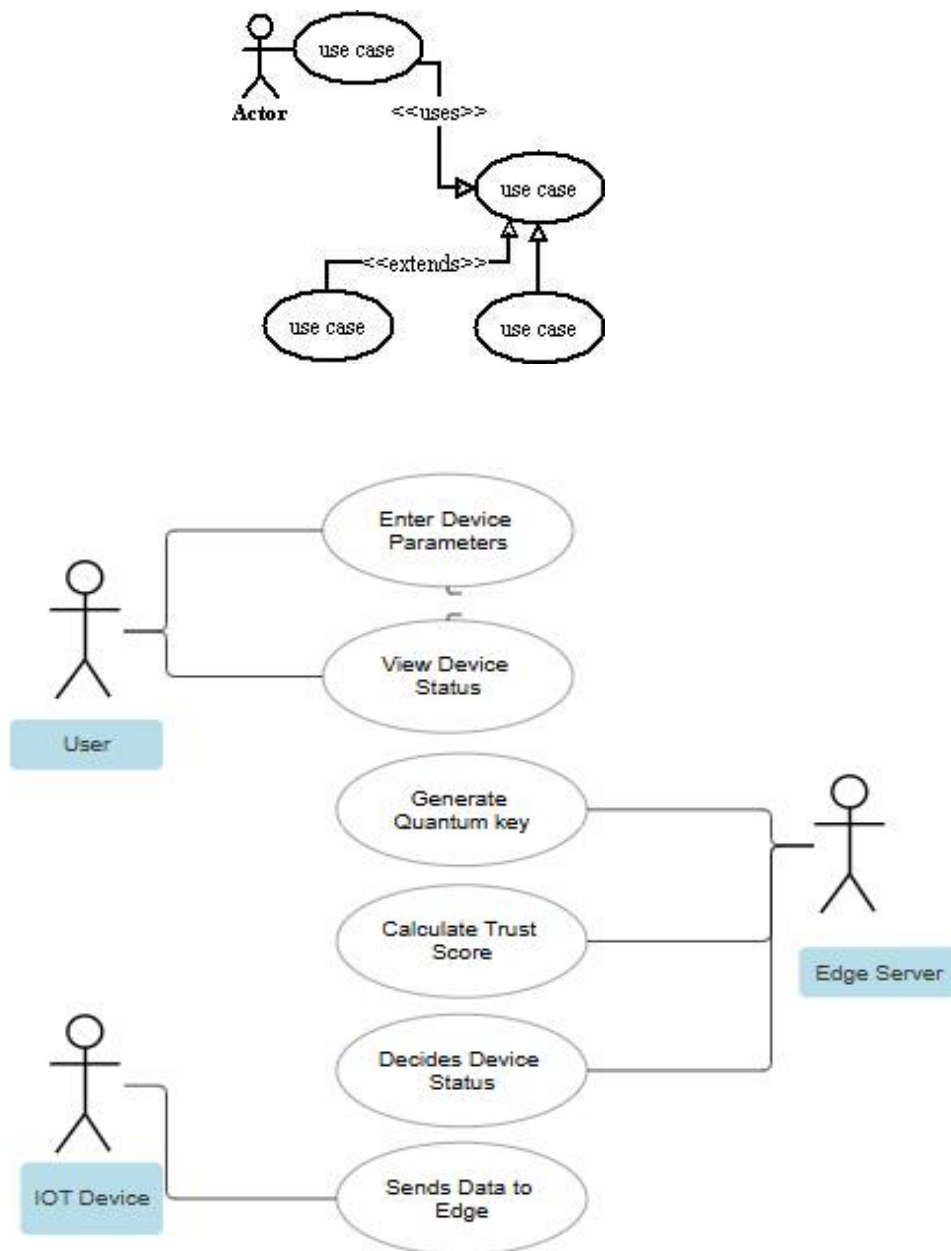


Fig 4.3: Use case Diagram of Trust-Aware Quantum-Assisted Security System for IoT Network.

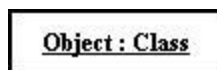
Sequence Diagram:

Sequence diagrams describe interactions among classes in terms of an exchange of messages over time.

Basic Sequence Diagram Symbols and Notations:

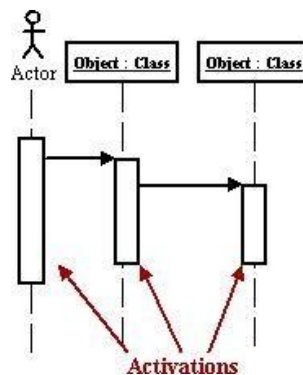
Class roles:

Class roles describe the way an object will behave in context. Use the UML object symbol to illustrate class roles, but don't list object attributes.

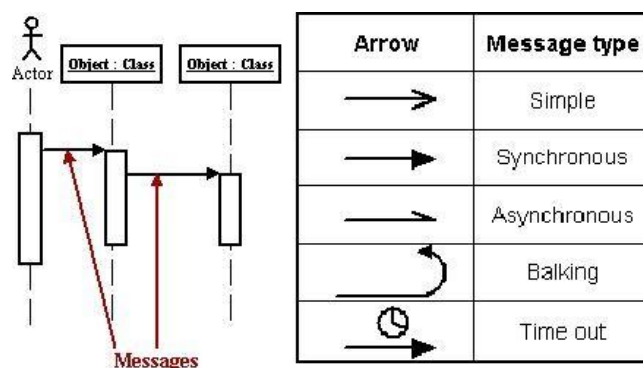


Activation:

Activation boxes represent the time an object needs to complete a task.



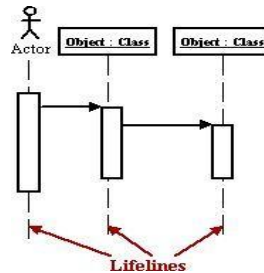
Messages:



Messages are arrows that represent communication between objects. Use half-arrowed lines to represent asynchronous messages. Asynchronous messages are sent from an object that will not wait for a response from the receiver before continuing its tasks

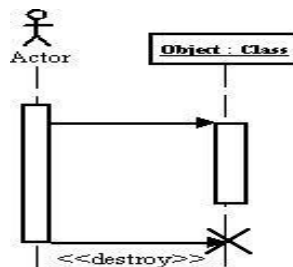
Lifelines:

Lifelines are vertical dashed lines that indicate the object's presence over time.



Destroying Objects:

Objects can be terminated early using an arrow labeled "<< destroy >>" that points to an X.



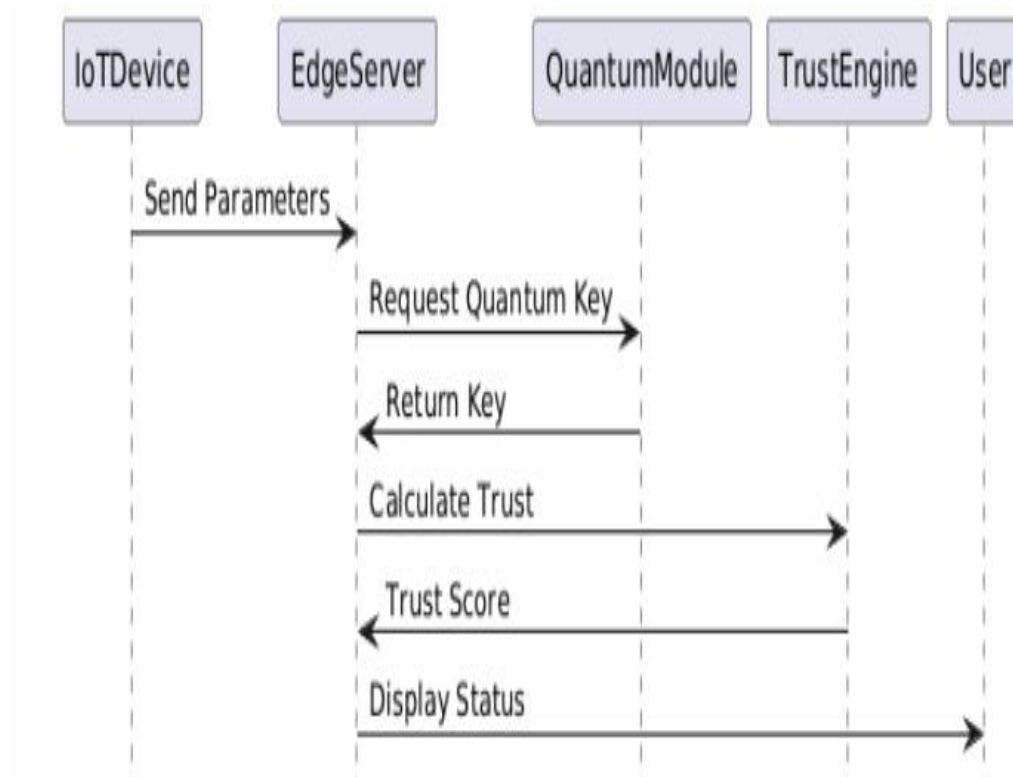


Fig 4.4: Sequence Diagram of Trust-Aware Quantum-Assisted Security System for IoT Network.

Activity Diagram:

An activity diagram illustrates the dynamic nature of a system by modelling the flow of control from activity to activity. An activity represents an operation on some class in the system that results in a change in the state of the system. Typically, activity diagrams are used to model workflow or business processes and internal operation. Because an activity diagram is a special kind of state chart diagram, it uses some of the same modelling conventions.

Basic Activity Diagram Symbols and Notations:

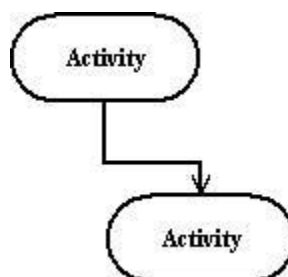
Action states:

Action states represent the non-interruptible actions of objects. You can draw an action state in Smart Draw using a rectangle with rounded corners.



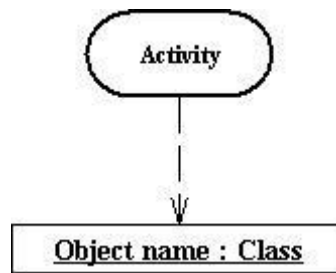
Action Flow:

Action flow arrows illustrate the relationships among action states.



Object Flow:

Object flow refers to the creation and modification of objects by activities. An object flow arrow from an action to an object means that the action creates or influences the object. An object flow arrow from an object to an action indicates that the action state uses the object.



Initial State:

A filled circle followed by an arrow represents the initial action state.



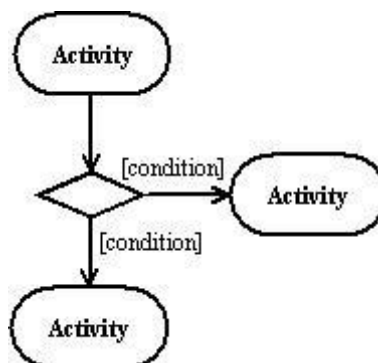
Final State:

An arrow pointing to a filled circle nested inside another circle represents the final action state.



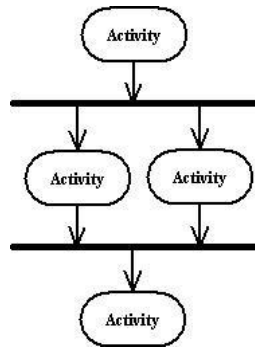
Branching:

A diamond represents a decision with alternate paths. The outgoing alternates should be labelled with a condition or guard expression. You can also label one of the paths "else."



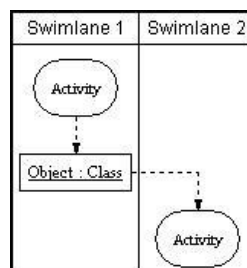
Synchronization:

A synchronization bar helps illustrate parallel transitions. Synchronization is also called forking and joining.



Swimlanes:

Swimlanes group related activities into one column.



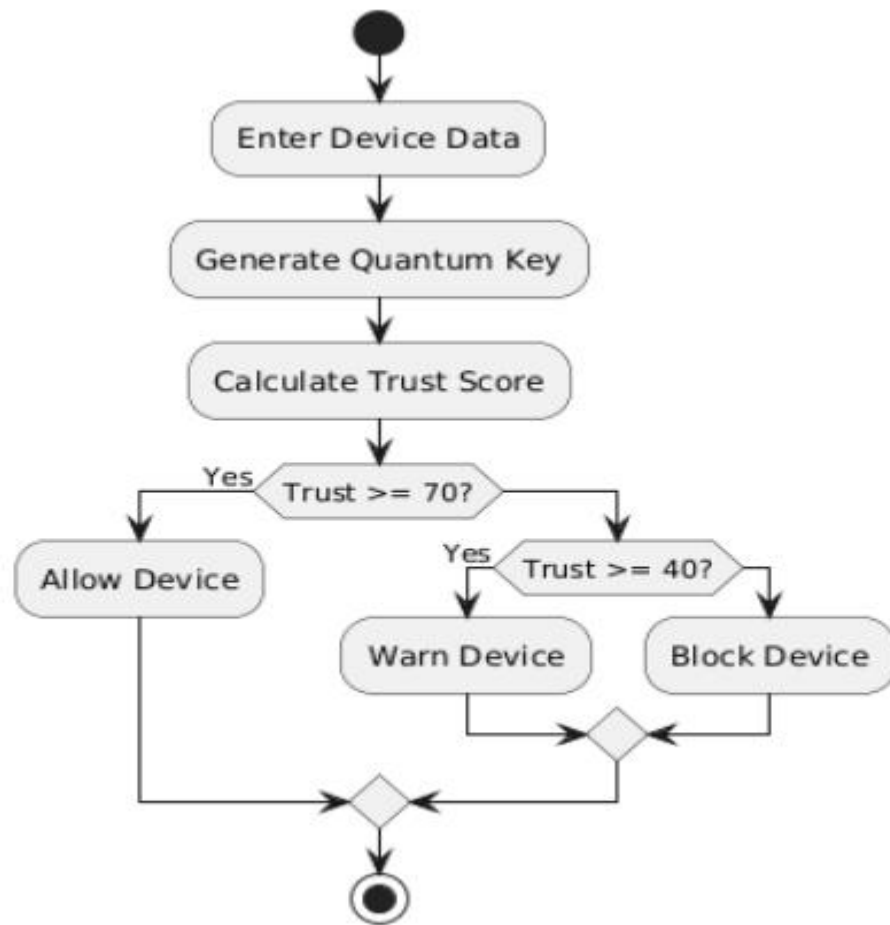


Fig 4.5: Activity Diagram of Trust-Aware Quantum-Assisted Security System for IoT Network.

System architecture:

System architecture is a conceptual model that describes the structure and behaviour of multiple components and subsystems.

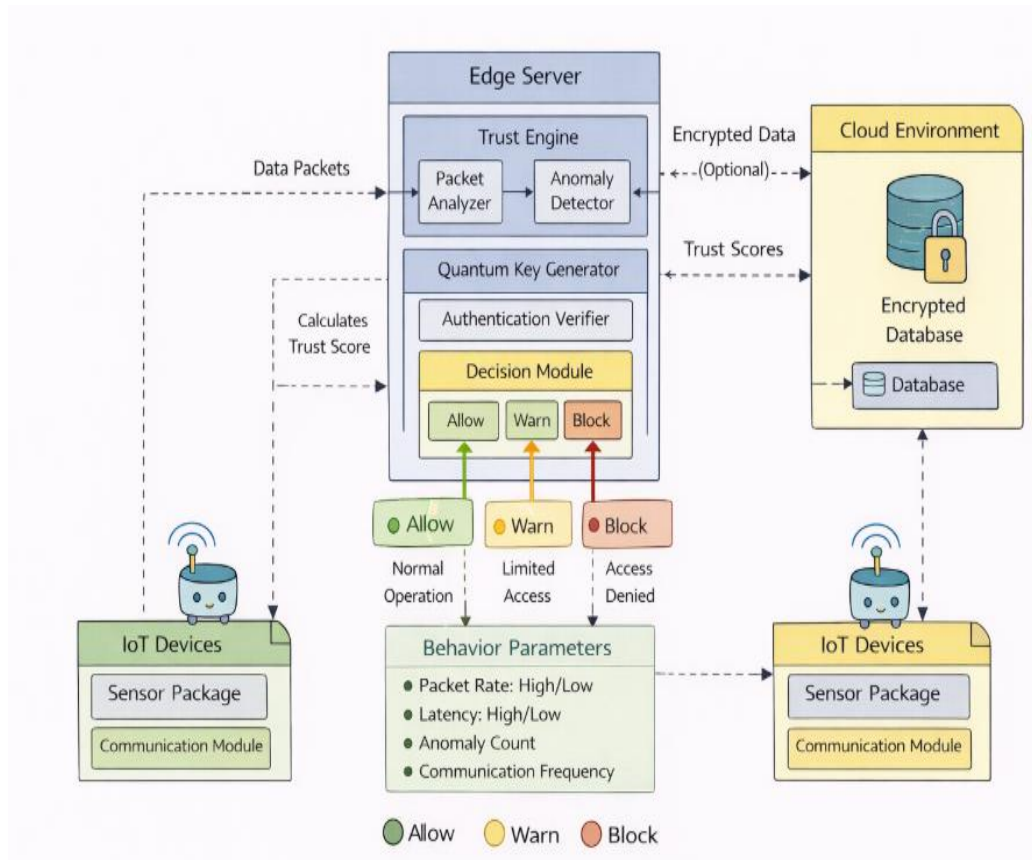


Fig 4.6: System architecture of Trust-Aware Quantum-Assisted Security System for IoT Network

5. SYSTEM IMPLEMENTATION

Introduction

System implementation is the phase in which the proposed Trust-Aware Quantum-Assisted IoT Security System is developed into a fully functional working model. This stage involves translating the system design into executable modules, integrating various components, and ensuring that the system performs efficiently under real-time conditions.

The implementation emphasizes:

- Secure communication among IoT devices
- Real-time monitoring and analysis
- Trust-based decision-making mechanisms
- Efficient processing using edge computing

The system is developed using a modular architecture, where each module is responsible for a specific functionality such as data acquisition, preprocessing, trust evaluation, encryption, and monitoring.

Data Acquisition

The system collects operational data from IoT devices deployed within the network. These devices may include sensors, embedded systems, or simulated nodes that continuously transmit their activity logs to the edge server.

Parameters Collected:

- Device Identification (Device ID)
- Packet Transmission Count
- Packet Delivery Ratio (PDR)
- Communication Frequency
- Response Time
- Error or Failure Rate

This data serves as the foundation for evaluating the reliability and behavior of each device in the network.

Data Preprocessing

Before performing trust evaluation, the collected data undergoes preprocessing to improve accuracy and consistency.

Key Preprocessing Steps:

- **Data Cleaning:** Removal of incomplete, inconsistent, or duplicate entries
- **Normalization:** Scaling of values to ensure uniform comparison
- **Noise Filtering:** Elimination of irregular spikes and anomalies
- **Data Structuring:** Organizing data into a standardized format

Normalization Formula:

$$X_{norm} = \frac{X - X_{min}}{X_{max} - X_{min}}$$

This ensures that all parameters contribute fairly during trust computation.

Trust Evaluation Module

The Trust Evaluation Module is the core component of the system, responsible for determining the reliability of each IoT device based on its behavior.

Parameters Considered:

- Packet Delivery Ratio (PDR)
- Communication Frequency (CF)
- Error Rate (ER)
- Response Time (RT)

Trust Score Calculation:

$$Trust = (w_1 \cdot PDR) + (w_2 \cdot CF) - (w_3 \cdot ER) - (w_4 \cdot RT)$$

Where:

- w_1, w_2, w_3, w_4 are weighting factors
- Positive parameters increase trust
- Negative parameters decrease trust

The weights can be adjusted based on system requirements and network conditions.

Decision-Making Mechanism

Based on the computed trust score, the system classifies devices into trusted or untrusted categories.

Decision Rule:

$$\text{If } Trust \geq T \Rightarrow \text{Trusted}$$

$$\text{If } Trust < T \Rightarrow \text{Untrusted}$$

Where T is a predefined threshold value.

System Actions:

- Trusted devices are allowed to communicate normally
- Untrusted devices are restricted or blocked
- Suspicious devices are flagged for monitoring

This mechanism helps in preventing malicious nodes from affecting the network.

Quantum-Assisted Security Mechanism

To enhance communication security, the system incorporates a quantum-inspired cryptographic approach for dynamic key generation.

Key Features:

- Random key generation for each session
- Frequent key updates to prevent attacks
- Secure encryption and decryption process

Encryption Process:

$$C = P \oplus K$$

Decryption Process:

$$P = C \oplus K$$

Where:

- P = Plain text
- C = Cipher text
- K = Secret key
- $\oplus$ = XOR operation

This ensures confidentiality and protects against unauthorized access.

Edge Server Implementation

An edge server is used to perform computations closer to the IoT devices, reducing dependency on centralized cloud systems.

Functions of Edge Server:

- Data collection from devices
- Preprocessing of incoming data
- Trust score computation

- Decision making
- Key generation and encryption

Advantages:

- Reduced latency
- Faster response time
- Improved system efficiency
- Lower bandwidth usage

System Modules

The system is divided into the following modules:

1. Data Collection Module

Handles gathering of device data in real-time.

2. Preprocessing Module

Cleans and prepares data for analysis.

3. Trust Evaluation Module

Calculates trust scores for each device.

4. Security Module

Implements encryption and key generation.

5. Monitoring Module

Displays system status and alerts.

Tools and Technologies Used

- **Python:** Core programming language for system development
- **Flask:** Web framework for backend implementation
- **HTML & CSS:** Frontend interface design

- **NumPy & Pandas:** Data processing and numerical computations

Monitoring and Alert System

A web-based dashboard is implemented to provide real-time system insights.

Features:

- Display of device trust scores
- Status classification (Trusted / Untrusted)
- Alert generation for suspicious activities

Alert Conditions:

- Significant drop in trust score
- High error rate
- Abnormal communication patterns

System Workflow

1. IoT devices generate and send data
2. Edge server receives data
3. Data preprocessing is performed
4. Trust score is calculated
5. Device is classified based on threshold
6. Secure key is generated
7. Data is encrypted and transmitted
8. Results are displayed on dashboard

6. RESULTS & DISCUSSION

```
1  /* General Page Styling */
2  body {
3      font-family: Arial, sans-serif;
4      background-color: #f4f6f8;
5      margin: 0;
6      padding: 0;
7      text-align: center;
8  }
9
10 /* Title */
11 h2 {
12     margin-top: 20px;
13     color: #333;
14 }
15
```

Fig 6.1:style.css

```
templates > <> index.html > <html> <head> <link>
1  <!DOCTYPE html>
2  <html>
3  <head>
4      <title>IoT Security Dashboard</title>
5      <link rel="stylesheet" href="{{ url_for('static', filename='style.css') }}">
6      <style>
7          body {
8              font-family: Arial;
9              text-align: center;
10         }
11         table {
```

Fig 6.2: index.html

```
app.py > ...
1  from flask import Flask, request, render_template
2
3  app = Flask(__name__)
4
5  devices = []
6
7  @app.route('/')
8  def home():
9      return render_template("index.html", devices=devices)
```

Fig 6.3: app.py

```

device_simulator.py > ...
1  import requests
2  import random
3  import time
4
5  devices = ["sensor_1", "sensor_2", "sensor_3"]
6
7  while True:
8      device = random.choice(devices)
9
10     packets_sent = random.randint(50, 100)

```

Fig 6.4: device_simulator.py

```

server.py > ...
1  from flask import Flask, request
2  import requests
3
4  app = Flask(__name__)
5
6  # Trust Calculation
7  def calculate_trust(data):
8      sent = data['packets_sent']
9      received = data['packets_received']
10
11     pdr = received / sent if sent != 0 else 0

```

Fig 6.5: server.py

```

requirements.txt
1  flask
2  requests

```

Fig 6.6: requirements.txt

```

PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS
PS C:\Users\sai sharath reddy\OneDrive\Desktop\mini project\iot> python device_simulator.py
[+] Data sent from sensor_2
[+] Data sent from sensor_3
[+] Data sent from sensor_3
[+] Data sent from sensor_3
[+] Data sent from sensor_3
[+] Data sent from sensor_3
[+] Data sent from sensor_3
[+] Data sent from sensor_1

```

Fig 6.7: Terminal 1

```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS python +
PS C:\Users\sai sharath reddy\OneDrive\Desktop\mini project\iot> python server.py
* Serving Flask app 'server'
* Debug mode: on
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on http://127.0.0.1:8000
Press CTRL+C to quit
* Restarting with watchdog (windowsapi)
* Debugger is active!
* Debugger PIN: 268-968-643

```

Fig 6.8: Terminal 2

```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS python +
PS C:\Users\sai sharath reddy\OneDrive\Desktop\mini project\iot> python app.py
* Serving Flask app 'app'
* Debug mode: on
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on http://127.0.0.1:5000
Press CTRL+C to quit
* Restarting with watchdog (windowsapi)

```

Fig 6.9:Terminal 3

IoT Trust Monitoring System		
Device ID	Trust Score	Status
sensor_3	0.363	✗ Untrusted
sensor_3	0.412	✗ Untrusted
sensor_1	0.565	✓ Trusted
sensor_1	0.093	✗ Untrusted
sensor_3	0.593	✓ Trusted
sensor_2	0.407	✗ Untrusted
sensor_3	0.474	✗ Untrusted
sensor_2	0.495	✗ Untrusted
sensor_1	0.02	✗ Untrusted
sensor_1	0.479	✗ Untrusted

Fig 6.7: output

7. SYSTEM TESTING

Testing is an important part of developing any security-based system. It helps ensure that the system works correctly, is free from major errors, and meets the required security standards. In IoT networks, proper testing is even more important because devices are highly distributed and vulnerable to attacks.

The proposed Trust-Aware Quantum-Assisted Security System is tested to verify secure communication, correct trust evaluation, and protection against malicious activities. Although this system is not based on deep learning, it still requires thorough validation to ensure reliability and robustness in real-world environments.

Testing in Traditional Systems vs Security Systems

In traditional software systems, testing mainly focuses on checking whether the output matches the expected behavior based on the written code.

In contrast, security systems like this project focus on:

- Detecting abnormal or malicious behavior
- Ensuring secure communication between devices
- Maintaining system stability under attacks

Testing here involves validating both functionality and security performance.

Testing Challenges

Some challenges faced while testing the system include:

- Dynamic nature of IoT networks (devices joining/leaving)
- Detecting unknown or new types of attacks
- Maintaining low latency while ensuring high security
- Handling large-scale device communication

Functional vs Security Evaluation

Testing and evaluation are not the same:

- Evaluation focuses on performance metrics like:
 - Detection accuracy
 - Trust score correctness
 - Response time
- Testing focuses on:
 - Whether malicious nodes are correctly identified
 - Whether secure communication is maintained
 - Whether the system behaves correctly under different scenarios

Both are required to ensure system reliability.

Types of Testing

1. Pre-Deployment Testing

These tests are performed before deploying the system:

- Verify node registration process
- Check encryption and decryption mechanisms
- Validate trust score calculation logic
- Ensure secure key generation
- Test communication between IoT devices

2. Post-Deployment Testing

These tests validate system behavior after deployment:

- Detection of malicious nodes
- Trust score variation with node behavior
- Secure data transmission under attack scenarios
- System response to abnormal activities
- Stability under continuous operation

Test Cases

Test No	Test Case	Expected Output	Actual Output	Result
1	System Initialization	System starts without errors	System initialized successfully	Pass
2	Node Registration	Node added to network	Node registered	Pass
3	Secure Communication	Data encrypted and transmitted	Secure transmission achieved	Pass
4	Trust Score Calculation	Correct trust value assigned	Accurate trust score generated	Pass
5	Malicious Node Detection	Malicious node identified	Node flagged successfully	Pass
6	Data Integrity Check	No data tampering	Data remains intact	Pass
7	Encryption/Decryption	Correct output	Data decrypted correctly	Pass
8	Network Performance	Low latency communication	Stable performance observed	Pass
9	Attack Simulation	Attack detected and prevented	System responded correctly	Pass

8. Mechanisms and Techniques

The proposed Trust-Aware Quantum-Assisted Security System for IoT Networks is designed using a combination of trust evaluation and secure communication mechanisms. The system focuses on lightweight and efficient algorithms that are suitable for IoT environments, where devices have limited computational and power resources.

To evaluate the reliability of devices in the network, the system uses a trust evaluation mechanism based on interaction history. In this approach, each IoT node maintains records of its communication behavior, such as successful data transmissions and failures. Based on these interactions, a trust score is calculated for every node. The trust value is dynamically updated as new interactions occur, allowing the system to adapt to changes in node behavior. If a node consistently shows abnormal or unreliable behavior, its trust score decreases, and it may be classified as malicious. This method helps in identifying compromised devices and prevents them from participating in network communication.

For securing data transmission, the system employs the Advanced Encryption Standard (AES) algorithm. AES is a symmetric key encryption technique that ensures confidentiality of data exchanged between IoT devices. Before transmission, the data is encrypted using a shared secret key, and at the receiver side, it is decrypted using the same key. The algorithm performs multiple rounds of substitution and permutation operations to transform plaintext into ciphertext, making it highly resistant to attacks. Due to its efficiency and low computational requirements, AES is widely used in IoT applications.

The system also incorporates a secure key generation and distribution mechanism inspired by quantum cryptographic principles. Instead of implementing full-scale quantum hardware, the model simulates quantum-assisted key exchange concepts to improve randomness and security in key generation. This approach helps in strengthening the encryption process by making the keys less predictable and more resistant to brute-force attacks. It enhances the overall security of the communication without introducing heavy computational overhead.

To monitor network behavior, a basic anomaly detection approach is used. The system continuously observes parameters such as packet delivery rate, communication frequency, and response time. Any deviation from normal behavior is treated as suspicious. When such anomalies are detected, the corresponding node's trust score is reduced, and appropriate actions are taken, such as restricting or

blocking the node. This ensures that the network remains secure even in the presence of malicious activities.

Overall, the combination of trust evaluation, AES-based encryption, quantum-inspired key generation, and anomaly detection provides a secure and efficient framework for protecting IoT networks. These algorithms work together to ensure data confidentiality, reliable communication, and effective detection of malicious nodes.

9. CONCLUSION

The proposed Trust-Aware Quantum-Assisted Security System for IoT Networks provides a secure and efficient approach to protect data communication in modern IoT environments. By combining trust evaluation mechanisms with quantum-assisted cryptographic techniques, the system addresses two major challenges in IoT networks: identifying malicious nodes and ensuring strong data security.

The trust model plays a key role in continuously monitoring node behavior and assigning trust scores, which helps in detecting and isolating untrusted or compromised devices. At the same time, the integration of quantum-assisted security techniques strengthens data protection against advanced cyber threats, including attacks that may break traditional encryption methods.

The system was able to maintain a good balance between security and performance, with minimal impact on latency and network throughput. Its lightweight design also makes it suitable for deployment in resource-constrained IoT environments such as smart homes, healthcare systems, and industrial networks.

However, the system has some limitations. The implementation of quantum-assisted techniques is currently dependent on simulations, as real quantum infrastructure is not widely available. Additionally, trust evaluation may take time in highly dynamic networks where device behavior changes frequently. There is also a slight increase in computational overhead compared to traditional security approaches.

For future improvements, the system can be enhanced by integrating real quantum communication technologies as they become more accessible. More advanced and adaptive trust models can be developed to improve detection accuracy in dynamic environments. The system can also be optimized for low-power IoT devices and extended to support large-scale applications such as smart cities and industrial IoT. Incorporating lightweight security mechanisms and hybrid architectures will further improve scalability and real-world usability.

Overall, this project demonstrates a practical and forward-looking approach to securing IoT networks by combining trust management with next-generation security techniques.

10.FUTURE ENHANCEMENT

Multi-Modal Security Integration: Incorporate multiple data sources such as network traffic patterns, device behavior logs, and environmental context along with trust metrics. By combining different types of inputs, the system can make more accurate and reliable security decisions in dynamic IoT environments.

Quantum Key Distribution (QKD) Integration: Extend the system to support real quantum-based encryption techniques like QKD instead of only quantum-inspired methods. This will significantly enhance security by making communication resistant to future quantum attacks.

AI-Based Adaptive Trust Models: Implement advanced machine learning models that continuously learn and update trust scores of IoT devices based on their behavior. This helps in detecting malicious nodes early and improving overall network reliability.

Blockchain-Based Trust Management: Integrate blockchain technology to store trust values and transaction records in a decentralized and tamper-proof manner. This improves transparency and eliminates single points of failure in the system.

Edge Computing Optimization: Deploy trust evaluation and security mechanisms at the edge level to reduce latency and improve real-time response. This is especially useful for large-scale IoT networks with limited bandwidth.

Ensemble Security Models: Combine multiple trust evaluation and anomaly detection models to improve accuracy. Ensemble approaches can provide more reliable results compared to a single model.

Real-Time Threat Detection and Response: Enhance the system with automated threat detection and immediate response mechanisms such as isolating compromised nodes or rerouting data securely.

Lightweight Security for Resource-Constrained Devices: Optimize algorithms to run efficiently on low-power IoT devices without affecting performance, ensuring wider adoption in real-world applications.

Interactive Monitoring Dashboard: Develop a user-friendly web or mobile interface for administrators to monitor trust levels, detect threats, and manage IoT devices in real time. This improves usability and control over the network.

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